

Guanxiong Luo, PhD in Computer Science, Machine Learning Researcher & Engineer

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Summary

Machine learning researcher and engineer with a PhD in Computer Science, specializing in generative models, computer vision, and inverse problems. Extensive experience developing diffusion models, autoregressive models, and self-supervised frameworks for image, video, and 3D generation, with first-author publications at NeurIPS and CVPR. Full-stack engineering across the ML stack: from-scratch GPU kernels (CUDA, Metal), distributed training (tensor/sequence parallelism, ZeRO), large-scale data pipelines, and model deployment. Strong mathematical foundation in numerical optimization, numerical linear algebra, and probabilistic methods, with hands-on proficiency in PyTorch, JAX, and TensorFlow.

Expertise

Generative models | Diffusion models | Autoregressive models | Computer vision | Deep learning | Inverse problems | Bayesian inference & uncertainty quantification | Numerical optimization | GPU acceleration (CUDA/Metal) | Distributed training | PyTorch | JAX | TensorFlow

Experiences

Employment

06/2026-present: Senior AI Algorithm Engineer (Contractor) at Apple, Munich

1. Develop computational imaging algorithms built on generative models (diffusion, autoregressive) for Apple product imaging pipelines.
2. Work with hardware and software engineering teams to optimize AI algorithms for Apple silicon, balancing model quality against compute and memory efficiency.

01/2026-06/2026: Independent Researcher — Quantitative Trading & ML

1. Authored in-depth technical deep-dives on Tensor Parallelism and ZeRO-1 Distributed Optimizer, with open-source reference implementations in PyTorch and NCCL collectives.
2. Researched quantitative trading strategies using probabilistic modeling, time-series analysis, and deep learning.

07/2025-12/2025: Postdoctoral Research Fellow at Luxembourg Institute of Health, Luxembourg

1. Developed state-of-the-art autoregressive diffusion models for 3D volume generation and temporal consistency in image sequences, with multi-GPU training on $4 \times$ NVIDIA A100 GPUs.
2. Designed and implemented ML systems for training pipelines, optimizing for efficient inference and scalability on HPC clusters.
3. Collaborated with cross-functional teams to productionize research innovations, delivering clean, well-tested code and reproducible experiments.

01/2020-10/2024: Research Scientist at University Medical Center Göttingen, Germany

1. Developed and implemented machine learning models and algorithms for large-scale image data analysis, with focus on computational imaging and computer vision.
2. Designed novel numerical algorithms for solving inverse problems using iterative optimization, Bayesian inference, and diffusion models, publishing at top venues (NeurIPS, CVPR).
3. Built end-to-end ML infrastructure: data pipelines, distributed multi-GPU training on HPC clusters, evaluation frameworks, and optimized inference engines.
4. Contributed to the open-source BART toolbox (C) used by the research community worldwide; mentored students and collaborated with teams across Europe, Asia, and North America.

09/2017-11/2019: Research Assistant at LKS Faculty of Medicine, University of Hong Kong

1. Applied numerical optimization and physics-informed modeling to develop AI-driven solutions for computational imaging.
2. Developed Bayesian estimation methods for image reconstruction using generative models as priors, laying the foundation for later work in uncertainty quantification.

Education

- **10/2020-11/2023:** PhD in Computer Science, University of Göttingen;
Advisor: Prof. Dr. Martin Uecker; Thesis: *Development of Advanced Generative Priors for MRI Reconstruction*
- **09/2017-10/2019:** M.Phil in Biomedical Engineering, University of Hong Kong;
Advisor: Prof. Dr. Cao Peng; Thesis: *The Application of Generative Networks in MR Image Reconstruction*
- **09/2013-07/2017:** B. Eng in Biomedical Engineering, Xi'an Jiaotong University;
Advisor: Prof. Dr. Junbo Duan; Final Project: *Fast MRI Using Compressed Sensing*.

Award & Honor

- **12/2023:** PhD Graduated with Magna cum Laude, University of Göttingen
- **2017-2019:** Postgraduate Scholarship awarded by The University of Hong Kong
- **06/2017:** Outstanding Graduate of Class 2017 awarded by Xi'an Jiaotong University
- **10/2015:** National Encouragement Scholarship awarded by Xi'an Jiaotong University
- **04/2015:** Meritorious Winner in American Mathematical Contest in Modeling (MCM)

Technical Skills & Open-Source Contributions

- **Programming:** Python, C/C++, CUDA, Metal, Shell scripting, Linux/Unix | **Deep Learning:** PyTorch, JAX, TensorFlow, MLX, TensorBoard
- **ML Systems:** Low-level GPU kernel optimization (CUDA, Metal), distributed training (DDP, tensor/sequence parallelism, ZeRO), TensorRT inference, HPC clusters (SLURM, MPI, Docker)
- **Research:** Diffusion models, autoregressive models, inverse problems, Bayesian inference & uncertainty quantification, numerical optimization, numerical linear algebra (FFT/NUFFT, SVD)
- **Selected Repositories:**
 - [aid](#) (PyTorch) — Autoregressive Image Diffusion (NeurIPS 2024)
 - [self-diffusion](#) (PyTorch) — Zero-shot diffusion framework (NeurIPS 2025, CVPR 2026)
 - [Qwen3.8-27B-MLX](#) (Python/C++, MLX) — 27B LLM from scratch, with fused Metal kernel and speculative decoding
 - [Minimal_Flash_Attention](#) (CUDA) — Flash Attention from scratch, 8 progressively optimized kernel variants
 - [tensor-parallel](#) (PyTorch/NCCL) — Tensor & sequence parallelism with communication-computation overlap
 - [mitax](#) (JAX/Flax) — Medical imaging toolbox with FFT/NUFFT operators and diffusion models
 - [TensorRT-Cpp-Example](#) (C++/CUDA) — TensorRT inference pipeline with ONNX export and FP16/INT8 optimization

Selected Projects

Self-Diffusion: Zero-Shot Generative Modeling for Inverse Problems

- **Role:** Lead Researcher and Developer | **Duration:** 2025/05-2026/01
- **Summary:** Developed self-diffusion, a framework that solves inverse problems with no training data, no pretrained model, and no external priors — alternating controlled noise injection with fitting an untrained network to the measurements alone, with theoretical analysis of the noise-modulated spectral bias that drives coarse-to-fine convergence. Extended to blind imaging (DeblurSDI), jointly estimating the image and an unknown degradation kernel using two coupled untrained networks, achieving 33.00 dB PSNR on FFHQ optical aberrations versus 26.31 dB for the next best method. [Published at NeurIPS 2025 and CVPR 2026.](#)

Autoregressive Image Diffusion (AID)

- **Role:** Lead Researcher and Developer | **Duration:** 2024/01-2024/06
- **Summary:** Built an autoregressive diffusion model combining causal attention (DiTBlock-based transformer architecture) with denoising diffusion to generate coherent image sequences, modelling inter-image dependencies rather than treating each frame independently. Demonstrated on video frame interpolation, temporal super-resolution, and 3D volume generation, with substantially reduced hallucinations compared to standard diffusion models. Supports both pixel-space and VQVAE-compressed latent-space operation; training parallelized across sequence positions on $4 \times$ NVIDIA A100 GPUs. [Published at NeurIPS 2024.](#)

27B Language Model From Scratch — Python and C++ (MLX)

- **Role:** Sole Author and Developer | **Duration:** 2026
- **Summary:** Implemented a 27B hybrid linear-attention language model from scratch against MLX, twice — first in Python, then ported to C++ via MLX's C++ API with no Python at runtime. Wrote the model, byte-level BPE tokenizer, quantizer, speculative decoder, and terminal assistant with no Transformers library and no network access. Includes a hand-written **fused Metal kernel** for the gated-delta recurrence (0.6% of FLOPs but $\sim 20\%$ of a decode step; $2.75 \times$ isolated, +4.8% end-to-end) and **lossless self-speculative decoding** via the model's own MTP head ($\sim 1.9 \times$ on structured output). Quantized a 55.6 GB bf16 checkpoint to 15.86 GB at 4-bit; roofline analysis showed decode is memory-bandwidth-bound, explaining why the C++ port matches Python within 5%.

Bayesian Inference with Joint Uncertainty Quantification

- **Role:** Lead Researcher and Developer | **Duration:** 2021/01-2022/12
- **Summary:** Developed a Bayesian inference framework using diffusion models and MCMC posterior sampling that produces pixel-wise uncertainty maps and 95% confidence intervals alongside each reconstruction, applicable to arbitrary linear inverse problems and noisy observations. Extensive evaluation of uncertainty calibration, noise modelling, and efficient sampling. [Published in Magnetic Resonance in Medicine \(top-ten cited article in 2023\).](#)

Production-Ready ML Infrastructure & Open-Source Tooling

- **Role:** Lead Researcher, Developer and Collaborator | **Duration:** 01/2020-03/2024
- **Summary:** Built end-to-end ML infrastructure for imaging research: data ingestion and quality control pipelines, multi-GPU distributed training with interruptible checkpointing, configuration-driven experiment management, evaluation frameworks, and TensorRT-optimized inference. Delivered production-grade command-line tools and Python libraries adopted by clinical researchers, and contributed to the open-source BART toolbox (C) used by the medical imaging community worldwide.

Publications

Conference Papers & Preprints

- [1] Yanlong Yang, **Guanxiong Luo***. *Self-diffusion Driven Blind Imaging*. Conference on Computer Vision and Pattern Recognition. ([Accepted by CVPR 2026, Equal contribution, arxiv:2510.27439](#))

- [2] **Guanxiong Luo**, Shoujin Huang, Yanlong Yang. *Self-diffusion for solving inverse problems*. Advances in Neural Information Processing Systems. Vol 39. Curran Associates, Inc.; 2025. ([Accepted by NeurIPS 2025](#), [arxiv:2510.21417](#))
- [3] **Guanxiong Luo**, Shoujin Huang, Martin Uecker. *Autoregressive Image Diffusion: Generation of Image Sequence and Application in MRI*. Advances in Neural Information Processing Systems. Vol 38. Curran Associates, Inc.; 2024. ([Accepted by NeurIPS 2024](#))
- [4] Shoujin Huang, **Guanxiong Luo**^{*}, Xi Wang, Ziran Chen, Yuwan Wang, Huaishui Yang, Pheng-Ann Heng, Lingyan Zhang, Mengye Lyu. *Noise Level Adaptive Diffusion Model for Robust Reconstruction of Accelerated MRI*. MICCAI 2024: 498-508. ([Equal contribution](#))
- [5] **Guanxiong Luo**, Mengmeng Kuang, Peng Cao. *Generalized Deep Learning-based Proximal Gradient Descent for MR Reconstruction*. International Conference on Artificial Intelligence in Medicine. Springer Nature Switzerland; 2023: 239-244.

Journal Articles

- [6] Shoujin Huang, **Guanxiong Luo**, Yuwan Wang, Kexin Yang, Lingyan Zhang, Jingzhe Liu, Hua Guo, Min Wang, Mengye Lyu. *Robust Simultaneous Multislice MRI Reconstruction Using Deep Generative Priors*, Med Image Anal. 2026; 108:103851.
- [7] **Guanxiong Luo**, Xiaoqing Wang, Moritz Blumenthal, Martin Schilling, Erik Hans Ulrich Rauf, Raviteja Kotikalapudi, Niels Focke, Martin Uecker *Generative Priors for MRI Reconstruction Trained from Magnitude-Only Images Using Phase Augmentation*, Phil. Trans. R. Soc. A. 2025; 383 (2299): 20240323. ([In theme issue on Generative modelling meets Bayesian inference: a new paradigm for inverse problems](#))
- [8] Zuojun Wang, **Guanxiong Luo**, Ye Li, Peng Cao. *Using a deep learning prior for accelerating hyperpolarized ¹³C MRSI on synthetic cancer datasets*. Magn Reson Med. 2024; 92(3): 945-955.
- [9] **Guanxiong Luo**, Moritz Blumenthal, Martin Heide, Martin Uecker. *Bayesian MRI Reconstruction with Joint Uncertainty Estimation Using Diffusion Priors*. Magn Reson Med. 2023; 90: 295-311. ([In collection of Machine Learning and AI in MRM, top-ten cited articles published in MRM in 2023, featured in the homepage of IBI at TU Graz](#))
- [10] Moritz Blumenthal, **Guanxiong Luo**, Martin Schilling, H. Christian M. Holme, Martin Uecker. *Deep, deep learning with BART*. Magn Reson Med. 2023; 89: 678-693. ([Top downloaded MRM article of 2022, featured in the homepage of IBI at TU Graz](#))
- [11] **Guanxiong Luo**, Na Zhao, Wenhao Jiang, Edward S. Hui, Peng Cao. *MRI reconstruction using deep Bayesian estimation*. Magn Reson Med. 2020; 84: 2246-2261.

Conference Abstracts

- [12] **Guanxiong Luo**, Moritz Blumenthal, Martin Heide, Martin Uecker *MRI Reconstruction Via Data-Driven Markov Chains With Joint Uncertainty Estimation*. Proceedings of the Annual Meeting of ISMRM. Vol 31. ISMRM; 2023: 0990. ([Oral presentation](#))
- [13] **Guanxiong Luo**, Martin Heide, Martin Uecker. *Using data-driven Markov chains for MRI reconstruction with Joint Uncertainty Estimation*. Proceedings of the Annual Meeting of ISMRM. Vol 30. ISMRM; 2022: 0298. ([Power Pitch](#))
- [14] **Guanxiong Luo**, Moritz Blumenthal, Xiaoqing Wang, Martin Uecker. *All you need are DICOM images*. Proceedings of the Annual Meeting of ISMRM. Vol 30. ISMRM; 2022: 1510.
- [15] **Guanxiong Luo**, Xiaoqing Wang, Volkert Roeloffs, Zhengguo Tan, Martin Uecker. *Joint estimation of coil sensitivities and image content using a deep image prior*. Proceedings of the Annual Meeting of ISMRM. Virtual Conference. Vol 29. ISMRM; 2021: 0280. ([Oral presentation](#))
- [16] **Guanxiong Luo**, Moritz Blumenthal, Martin Uecker. *Using data-driven image priors for image reconstruction with BART*. Proceedings of the Annual Meeting of ISMRM. Virtual Conference. Vol 29. ISMRM; 2021: 1756.
- [17] **Guanxiong Luo**, Cao Peng. *MRI Reconstruction Using Deep Bayesian Inference*. Proceedings of the Annual Meeting of ISMRM. Virtual Conference. Vol 28. ISMRM; 2020: 0996. ([Oral presentation](#))